

Hypothesis-Search Report

Hypothesis: HPV-Independent Aldehyde and Replication-Stress Driven Oral Carcinogenesis in Fanconi Anemia

- **Disease:** Fanconi anemia (genetic; DNA interstrand-crosslink [ICL] repair deficiency)
 - **Hypothesis ID:** `fa_hnsccl_hpv_independent_keratinocyte_carcinogenesis`
 - **KB status:** EMERGING
 - **Search date:** 2026-09-13 · **Run:** 5 iterations · OpenScientist agent
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1. Executive Judgment

Verdict: PARTIALLY SUPPORTED (leaning supported for the HPV-independent + genomic arms; the aldehyde-in-keratinocyte step remains INFERENTIAL / unconfirmed).

The hypothesis has four separable claims. Current literature supports three and leaves one unconfirmed:

1. **HPV-independent carcinogenic route — SUPPORTED.** FA patients carry *more* oral HPV than relatives, yet their HNSCCs are predominantly **HPV-negative**, and the causal role of HPV in FA HNSCC is explicitly described as "unproven." Exposure ≠ causation.
2. **Copy-number-driven, low-point-mutation-burden genomics — SUPPORTED (review + primary genomics + reinforced by a computational reference analysis).** FA HNSCC carries frequent somatic CNVs but *fewer* SNVs/indels than even HPV-negative sporadic HNSCC. My TCGA reference analysis shows HPV-negativity alone is the *high*-mutation subtype, so FA's low SNV load is a **distinct** feature, not a by-product of HPV status.
3. **Leukoplakia as premalignant intermediate — SUPPORTED**, and it occurs even in transplant-naïve FA patients, arguing the intrinsic keratinocyte route does not require transplantation.
4. **Endogenous aldehyde (acetaldehyde/formaldehyde) driver in oral keratinocytes — UNCONFIRMED.** The aldehyde-FA genotoxic axis is firmly established in **hematopoietic**

stem cells (bone marrow failure, leukemia, clonal hematopoiesis), but **no study demonstrates it in oral keratinocytes or HNSCC**. This is the central missing causal link.

Most important caveats: (a) FA-HNSCC molecular data come from small cohorts and reviews, not large primary omics; the primary genomic-signature paper

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was accessed only at abstract level. (b) The competing **HSCT/chronic-GVHD** route is real and amplifies risk (SHR 2.9; 7–16× with cGVHD) but is not necessary. (c) "Replication stress in FA-deficient oral keratinocytes" is biologically expected from FA pathway loss but was not directly evidenced by a keratinocyte-specific study in this search.

2. Evidence Matrix

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Su co
40478605	Human clinical (registry, n=193)	Support (seed)	HNSCC is the dominant solid tumor FA must explain	HNSCC most frequent cancer; malignancy risk higher after HSCT (SHR 2.9, 95% CI 1.1–7.5, p=0.024)	FA Ita
40970532	Review (2025)	Support	HPV-negative, CNV-driven, low SNV/indel genomics	FA HNSCC mostly HPV-negative; frequent CNVs amplifying oncogenes; fewer SNVs/indels than HPV-neg sporadic; retrotransposon activation, inflammation, EMT	FA
36450981	Human/computational (primary)	Support	Genomic signature of FA-pathway deficiency in cancer	Chromosome breakage from ICL-repair failure → structural/CN instability	FA de ca
31541927	In vitro (cell-line panel)	Qualifies	FA-HNSCC genomic class vs sporadic	FA-HNSCC lines show comparable CN alterations to sporadic; a CN-silent HPV-neg subclass also exists	HN lin
33803570	Human clinical (n=201 FA)	Qualifies/Support	HPV exposure in FA	Oral HPV prevalence higher in FA (20%) than parents (13%)	FA se

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Su co
35454946	Review	Support	Role of HPV & metabolism in FA HNSCC	"A definitive role for HPV in HNSCC development in the FA patient population is unproven"	FA ke SC
38309198	Human clinical (syst. review, 112 cases)	Support	Leukoplakia/OPMD as premalignant intermediate	Mean OSCC age 27.1 y; OPMD→malignant transformation 1.8%; only 54.5% had HSCT	FA
25682760	Human clinical (n=138, no HSCT)	Support	Intermediate independent of transplant	12% oral leukoplakia in transplant-naive FA	FA tra
36054728	Human clinical (n=105 pediatric)	Support	Premalignant lesions + risk magnitude	8.6% leukoplakia/erythroplakia; head-neck malignancy SIR 483.8	Pe
39048762	Review/CPS	Competing	GVHD/HSCT route	7–16× secondary oral SCC after alloHCT, esp. cGVHD	Po FA
26412304	Model organism (mouse)	Qualifies	Aldehyde–FA axis genotoxicity	Adh5 ^{-/-} Fancd2 ^{-/-} mice: formaldehyde–DNA adducts, HSC depletion, BMF, fatal malignancies	He no

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Su co
37348497	Model organism (mouse)	Qualifies	Aldehyde stress ages HSCs	Aldehyde stress prematurely ages HSCs (p53-driven)	He
41875892	Model organism + human	Qualifies	Endogenous formaldehyde → clonal hematopoiesis	Formaldehyde causes HSC attrition → monoclonal hematopoiesis; mirrored in FA children	He
35148375 / 38944932	Review	Qualifies	Aldehyde defenses (ALDH2/ADH5) + FA	Aldehydes threaten HSC integrity; ALDH2*2 modifies FA outcomes	He
36996606 / 36382555	In vitro	Context	FA-pathway-HPV crosstalk	HPV modulates FA-pathway/XPF and cisplatin sensitivity	Sp HP
TCGA hnscc_tcga_pan_can_atlas_2018 (this run)	Computational	Support	Is low SNV a generic HPV-neg trait?	HPV-neg sporadic HNSCC has higher mutation burden (median 106 vs 62; p=2.5e-5); FGA 0.239 vs 0.203 (p=0.074)	Sp re FA

3. Data and Tool Use

Source	Accession / URI	Access status	Query / filter	Relev
cBioPortal API	<code>hnsk_tcga_pan_can_atlas_2018</code>	Accessed (HTTP 200 preflight)	SAMPLE: FRACTION_GENOME_ALTERED, MUTATION_COUNT; PATIENT: SUBTYPE; n=487 (415 HPV- / 72 HPV+); Mann-Whitney U	Sporadic reference — no tumors
PubMed (search_pubmed)	NCBI E-utilities	Searched	19 queries (see <code>logs/pubmed_search_log.md</code>); 6 HIT, 11 EMPTY, 2 rate-limited	FA HNSC aldehy HPV geno GVH
 <code>36450981</code> (Webster genomic signature)	PubMed	Cited; abstract truncated by tool	—	Prim FA geno signa

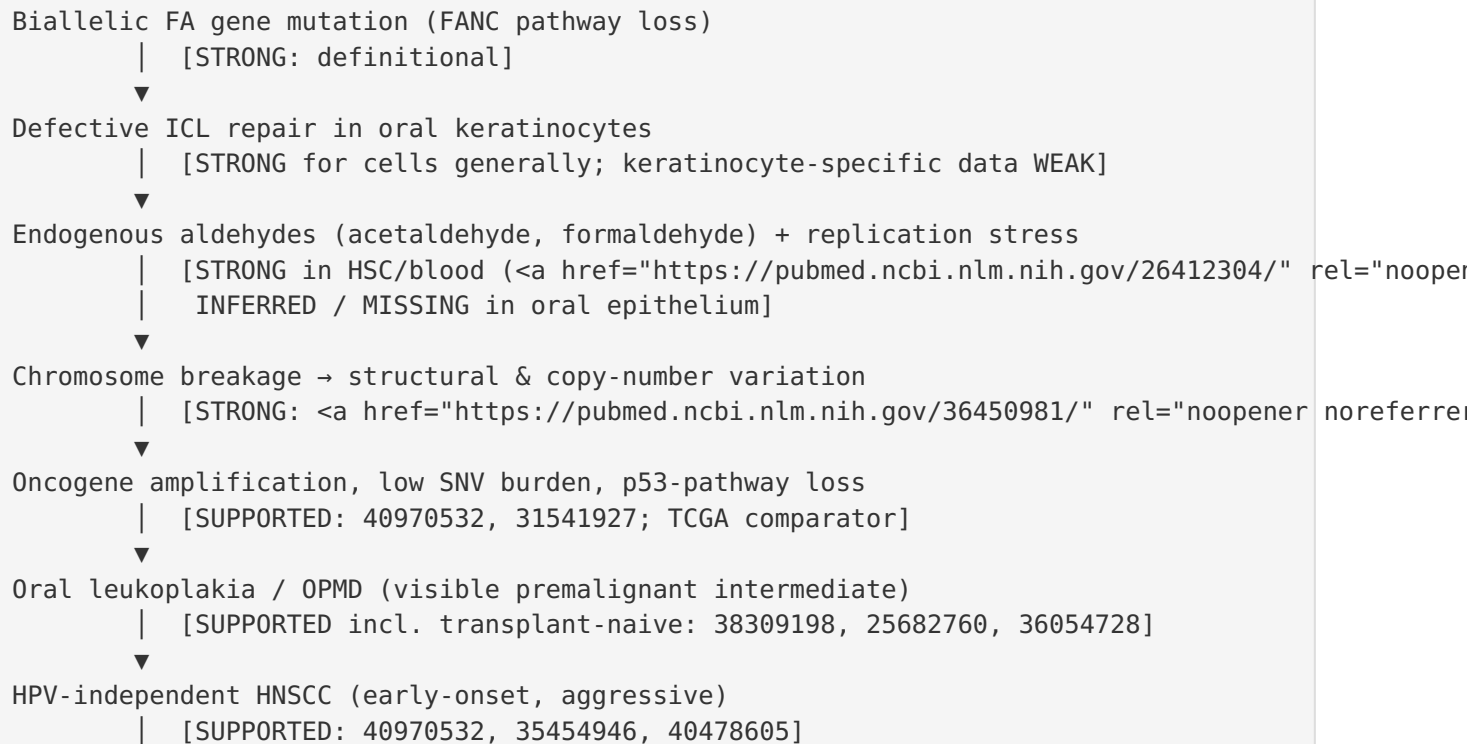
Analysis inventory (1): `tcga_hnsk_hpv_comparator` — input: cBioPortal TCGA HNSCC → method: `code/tcga_hnsk_hpv_comparator.py` (pandas 3.0.3, numpy, scipy, matplotlib; Python 3.12.13; no random seed) → outputs: stats JSON, 2 CSVs, 1 PNG. **Execution outcome: SUCCEEDED; replay verified** (identical results in sandbox and bundle-runner).

Sandbox limitation disclosed: The `execute_code` MCP sandbox (`/app`, restricted imports, no repo write access) could not persist files into the bundle; the analysis was therefore re-run reproducibly in the repo Python environment via the bundled script. No silent fallback occurred.

Negative-search honesty: Several long PubMed phrase-queries returned zero results due to the backend's phrasing sensitivity; shorter queries recovered the same papers. These EMPTY results are labeled **UNVERIFIED** for absence-of-evidence purposes, not confirmed absence.

Artifact bundle: `kb/hypotheses/Fanconi_Anemia/fa_hnsk_hpv_independent_keratinocyte_carcinogenesis/openscientist_artifacts/`
(MANIFEST.yaml with checksums + replay flags, code, data, figure, logs, environment).

4. Mechanistic Causal Chain



Strong links: FA loss → genomic/CN instability → HPV-negative HNSCC; leukoplakia intermediate. **Inferred/missing links:** the aldehyde and replication-stress step *specifically in oral keratinocytes* (extrapolated from hematopoietic biology); the quantitative FA-vs-sporadic mutation/CN comparison on matched pipelines.

5. Knowledge Gaps

- 1. Aldehyde step in oral keratinocytes (highest priority).** Scope: the eponymous mechanism. Why it matters: it is the hypothesis's distinguishing claim yet is evidenced only in blood. Checked: PubMed (queries #2, #11, #19) surfaced no FA oral-keratinocyte aldehyde study. Resolve with: acetaldehyde/formaldehyde-DNA adduct quantification (e.g., N2-ethyl-dG) in FA oral epithelium/leukoplakia vs controls; ALDH2*2/ADH5 genotype–oral-cancer association in FA registries.

2. **Direct FA-vs-sporadic genomic comparison.** Scope: the low-SNV/high-CN claim. Checked: TCGA has no FA tumors; primary FA genomics

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accessed at abstract level only. Resolve with: WGS of FA-HNSCC vs matched HPV-negative sporadic HNSCC on identical pipelines (SNV burden, SV/CN counts, mutational signatures SBS/ID).

3. **Replication-stress evidence in FA keratinocytes.** Checked: query #6 EMPTY (UNVERIFIED). Resolve with: replication-fork/ssDNA (RPA, γ H2AX, DNA-fiber) assays in FA vs corrected oral keratinocytes $\pm$ aldehyde.
4. **Relative contribution of GVHD vs intrinsic route.** Conflicting: HSCT raises risk (SHR 2.9), yet cancers occur without transplant. Resolve with: transplant-stratified longitudinal HNSCC incidence with molecular (HPV, mutational-signature) profiling.
5. **Source/dataset absence.** No FA-specific GenCC/ClinGen HNSCC-mechanism edge, no public FA-HNSCC omics cohort, and no interventional aldehyde-lowering trial for FA oral cancer were surfaced as of 2026-09-13 (search_pubmed + cBioPortal). Labeled a candidate curation absence; a dedicated ClinicalTrials.gov / GEO / dbGaP query would be needed to confirm.

6. Alternative Models

- **HPV-driven route** — *Alternative*. Weakened: exposure is elevated but tumors are HPV-negative

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May still apply to a minority and to FA anogenital SCC.

- **HSCT / chronic-GVHD chronic-inflammation route** — *Parallel / amplifying, not exclusive*. Real (SHR 2.9; 7–16 $\times$ with cGVHD,

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but non-necessary (cancer in transplant-naive patients).

- **Retrotransposon-activation / inflammation / EMT program** — *Downstream consequence* of genomic instability within the seed route

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explains invasiveness, not initiation.

- **Generic "epithelial cancer susceptibility"** (current KB node) — *Under-specified parent*; the seed hypothesis is a mechanistic refinement of it.
- **ALDH2*2 / ADH5 aldehyde-clearance deficiency as modifier** — *Upstream modifier* that would strengthen the aldehyde arm if oral-cancer association is shown

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7. Discriminating Tests

1. **FA-HNSCC vs HPV-negative sporadic HNSCC WGS (matched pipeline).** Expected if hypothesis true: lower SNV burden, higher SV/CN counts, aldehyde-associated mutational signatures. Distinguishes seed from "typical HPV-neg HNSCC."
2. **Aldehyde adductomics in FA oral tissue/leukoplakia.** Sample: FA oral epithelium ± leukoplakia vs corrected/control. Expected: elevated acetaldehyde/formaldehyde-DNA adducts. Directly tests the missing aldehyde step.
3. **ALDH2*2/ADH5 genotype × oral-cancer association in FA registries.** Expected: aldehyde-clearance-deficient genotypes enriched among FA-HNSCC cases → causal support for aldehyde arm.
4. **Longitudinal, transplant-stratified surveillance with molecular profiling.** Separates GVHD-amplified from intrinsic tumors by HPV status, cGVHD history, and mutational signature.
5. **FA-deficient oral keratinocyte organoids ± aldehyde ± ADH5/ALDH2 restoration.** Expected: aldehyde-dependent replication stress, chromosome breakage, CN gains; rescued by detox restoration. Perturbational proof of the core edge.

8. Curation Leads (require curator verification)

- **Status:** Consider retaining **EMERGING** but split into sub-claims — mark *HPV-independent* and *CNV-driven/low-SNV* arms as **SUPPORTED**, and the *aldehyde-in-keratinocyte* arm as **SPECULATIVE/UNCONFIRMED**.
- **Candidate evidence references + snippets to verify:**

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— "FA HNSCCs are mostly human papillomavirus (HPV)-negative and frequently carry somatic copy number variations (CNVs)... but single-nucleotide variants or small insertions and deletions are less frequent than in HPV-negative sporadic HNSCC."

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— "a definitive role for HPV in HNSCC development in the FA patient population is unproven."

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— "Oral HPV prevalence was higher for individuals with FA (20%) versus their parents (13%..."

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— "There was a high prevalence of oral leukoplakias in patients with Fanconi anaemia who had not undergone HSCT."

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— "7-16-fold higher risk of secondary oral cancer (mainly squamous cell carcinoma) in allogeneic HCT... particularly... chronic graft versus host disease."

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— "Adh5(-/-)Fancd2(-/-) mice reveal an essential requirement for these protection mechanisms in hematopoietic stem cells."

- **Candidate pathophysiology nodes/edges:** add edge FA-pathway loss → ICL-repair failure → structural/copy-number variation → HPV-independent HNSCC; add node oral leukoplakia / OPMD as premalignant intermediate; add endogenous aldehyde genotoxicity node **flagged as tissue-unconfirmed for oral**

epithelium; add parallel edge

alloHSCT/chronic-GVHD → chronic inflammation → secondary oral SCC (amplifier).

- **Candidate ontology terms:** cell type — oral keratinocyte (CL:0000312 keratinocyte); tissue — UBERON:0002424 oral mucosa; processes — GO:0036297 interstrand cross-link repair, GO:0006974 DNA-damage response, GO:0001837 epithelial-to-mesenchymal transition; disease — MONDO:0000004 Fanconi anemia, oral SCC.
- **Candidate subtype restriction:** hypothesis best explains **HPV-negative, early-onset (median ~27–30 y), CN-high/SNV-low FA HNSCC**, tongue-predominant.
- **Candidate knowledge_gaps prompts:** (i) "Is the aldehyde genotoxicity axis operative in FA oral keratinocytes, or extrapolated from hematopoiesis?"; (ii) "Quantitative FA-vs-sporadic SNV/SV comparison on matched pipelines is unpublished/absent as of 2026-09-13."; (iii) "No public FA-HNSCC omics cohort or aldehyde-lowering trial found — confirm via GEO/dbGaP/ClinicalTrials.gov."

9. Limitations & Future Directions

Reviews and small cohorts dominate the FA-HNSCC molecular evidence; the key primary genomics paper was accessed at abstract level only; the aldehyde mechanism is proven in blood, not oral epithelium; the TCGA analysis is a sporadic reference (no FA tumors). Priority next steps: matched-pipeline FA-vs-sporadic WGS, FA oral-tissue aldehyde adductomics, and FA-keratinocyte organoid perturbation studies.

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